



Zoology Legends

Z00507

Current Final TERM Spring 2023

TAAHA KHAN

Admin of Zoology Legend

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MCQS Spring 2023

1. Fauna of temperate deciduous forest **all (rodents, deer and bison.)**
2. Fauna of arid sub tropical ecosystem include **all (Indian robin, black myna, partridge)**
3. **Both (Littoral , Inter tidal Zone)** Specifically, Indus mouth and bays along Makran Coast being characterized by Mangroves.
4. Indian Thar deserts are an extensions of **Sahara** deserts through Arabian and Persian deserts.
5. Lotic freshwater zones are **2**.
6. Tropical grasslands are commonly called **Savannas**.
7. Grassland occupy about **20%** of the land on earth's surface.
8. The Chinese zodiac includes sheep among **the 12** animals.
9. Which ecosystem is known as the land of midnight of sun? **Tundra ecosystem**
10. **Gamma rays** cause damage to cell structure, cancer.
11. Coniferous forest are characterized by **both (conifers gymnosperms)**.
12. The surface layer of water that is constantly mixed by wind and waves and is warmed by sun , from late spring to late fall. **Epilimnion**
13. **Four** climatic zones of Pakistan.
14. K-2 **(8611m)**
15. National bird of Pakistan **choukor**
16. Choukor is present in the **Dry temperate forests**
17. Nitrogen fixation done by enzyme **nitrogenase**
18. Paracitism sign is **(+/-)**



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19. Heat always flows spontaneously from **hotter to colder bodies** , and never the reverse.
20. **Denitrification** is the process in which the nitrogen compounds makes its way back into the atmosphere.
21. The snow leopard is still occasionally called **ounce** .
22. Animals living at high altitudes **all (Snowleopard, snow pigeon, snow Partridge).**
23. Frequency of sound human can hear are **ultrasonic**
24. **Light trespass** is light being cast where it is not wanted or needed, such as light from a streetlight or a floodlight that illuminates a neighbor's bedroom at night making it difficult to sleep.
25. **70%** of Earth's surface is seawater.
26. Normal rainwater (**pH- 5.3-6**)
27. **Torpor is** a mechanism that is under physiological control.
28. Oil rich country according to per Capita ecological footprint **United States Emirates.**
29. Aquatic animals live in the pH of **5** .
30. **Ecology is** the scientific study of the interactions that determine the distribution and abundance of organisms in an environment.
31. Nitrogen in atmosphere is **78%**
32. Which is not a aquatic autotroph **eucalyptus**
33. **Biological diversity** is variety of all life forms on earth.
34. Deforestation **increase** CO₂ concentration in the atmosphere.
35. The average temperature of earth surface is increasing day by day due to increase in concentration of greenhouse gases in the atmosphere. **Global warming**
36. Ecology was firstly used in **1869**.



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37. Word ecology was first used by **Ernest Haeckel**
38. Types of lotic water **3**.
39. Types of pollution **.2**
40. Competition sign **(-/-)**
41. 85% of **phosphorus** present in bones and teeth.
42. Ozone absorbs ...**99%**..... Percent of UV rays.
43. A condition in which animals become inactive usually in cold environments for their **survival is Hibernation** .
44. Sunlight is a **renewable** source.
45. Waste mining generated by **3** ways
46. Waste is type of **land** pollution.
47. How many types of water pollution **5..?**
48. **cosmic rays**. rays are produced by Stars in Galaxy.
49. Scientific name of **snow leopard is Panthera uncia**.
50. Snow leopard found in Ecosystem **Sub Alpine and birch forest**.
51. The factors that brings undesirable change in environment are called **pollutants** .
52. Conifer deciduous forest has annual rain fall.... **200cm**
53. Who describes biological diversity first... **Robert Whittaker**.
54. Planting of trees is called as **Afforestation** .
55. pH of acid rain is **4**.
56. Height of K-2 ... **8611** .
57. world's 8th highest peak in Pakistan.... , **Nanga parbat**
58. The ecosystem where chukoor found..in **.Sub Alpine and Birch Forest**.
59. sign of bravery and competition... **Markhor** .
60. Unwanted Sound is called **noise** pollution.
61. Types of mining **.3**
62. **Light trespass** is light being cast where it is not wanted or needed, such as light from a streetlight or a floodlight that illuminates a neighbor's bedroom at night making it difficult to sleep.



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- 63. concentration **of CO2** **in**crease in global warming.
- 64. An increase in wildfires, due to polluted areas often becoming very **dry**
- 65. Milling operations- through various technologies Waste material is called **tailing**
- 66. 70% of Earth's surface is **seawater**
- 67. More than 90% of freshwater is **groundwater**
- 68. A large regional community is known as **biome**.



ZOOLOGY
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Repeated Questions Spring 2023

- ***_What is the scope of ecology? (3)**

Scope of ecology

- i) all organisms living on Earth
- ii) all physical and chemical surroundings.
- iii) for this reason, the field is usually divided into different levels of study including: organismal ecology, population ecology, community ecology and ecosystem ecology.

- **How ozone rate is described in stratosphere and troposphere?(3)**

Stratospheric ozone:

In the first step, solar ultraviolet radiation breaks apart one oxygen molecule (O_2) to produce two oxygen atoms. In the second step, each of these highly reactive atoms combines with an oxygen molecule to produce an ozone molecule (O_3).

Tropospheric ozone:

Nitrogen compounds released from automobiles is broken down by sunlight into nitrogen oxide (NO_2). It is then converted in to nitrogen monoxide (NO), releasing an oxygen molecule that combines with oxygen (O_2) in the atmosphere to form Ozone (O_3).

- **What is reproductive potential? Which factors influence reproductive potential? (5)**

Reproductive potential

Ability of individuals to reproduce under ideal or optimum conditions. Exponential growth cannot occur indefinitely.

Factors influencing reproductive potential

- ☐ Number of offspring produced
- ☐ The likelihood of survival to reproductive age
- ☐ Duration of the reproductive period
- ☐ Length of time it takes to reach maturity

- **Types of noise pollution.(5)**

Types of Noise Pollution

Following are the three types of pollution:

- i) Transport Noise- consists of traffic noise with the increase in the number of vehicles. leads to deafening of older people, headache, hypertension.



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- ii) Neighborhood Noise- household utensils, musical instruments, transistors, loudspeakers, etc.
- iii) Industrial Noise-heavy industrial machines, industrial noise pollution damages the hearing ability to around 20%

- **Threats to snow leopard._ (5)_**

Threats to snow leopards:

- ☐ Habitat degradation and fragmentation
- ☐ Reduction of natural prey due to illegal hunting
- ☐ Retribution killing for livestock loss
- ☐ War and related military activities

Effects of Radiations

Short-range effects

- Burns
- Impaired metabolism
- Dead tissues
- Death of the organisms.
- Selectively deposit

Long-range effects

- Mutations
- Increased incidence of tumors and cancer
- Shortening of life-span
- Developmental changes.
- Selectively deposit

Some species of animals and plants preferentially accumulate specific radioactive materials. For example, oysters deposit ^{65}Zn , fish accumulate ^{55}Fe , marine animals.

How lakes ground water are formed and their causes or factors?.

Lakes are large natural bodies of standing freshwater formed when precipitation, runoff, or groundwater seepage fills depressions in the earth's surface. Lake formation

Factors that are vital in lake formation are

- Glaciation
- Crustal displacement
- Volcanic activity

National animal of Pakistan

The markhor-national animal of Pakistan.

☐ Pakistan's national animal, "Markhor" symbol of bravery and sharp intelligence. ☐ Haven't any fear during prey and jump from the peak of the mountain to the base.

☐ It was one of the 72 animals featured on the WWF Conservation Coin Collection in 1976.



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Markhor is the National animal of Pakistan.

Scientific name: *Capra falconeri falconeri*

Capra- Genus of wild goat:

Species of Markhor

Distinct five subspecies- based on horn shape

☐ *Capra falconeri falconeri*- Astor Markhor- (flare-horned markhor) confined to Gilgit

☐ *Capra falconeri cashmirensis*- Kashmir Markhor- chitral, swat and Azad Kashmir.

Capra falconeri megaceros- Kabul Markhor- Malakand, swat Chitral

☐ *Capra falconeri jerdoni*- Suleman Markhor- mountain ranges of north and east of Quetta: October 29, 1988-PPO

☐ *Capra falconeri chiltanesis*- Chiltan Markhor- three hill ranges (Chiltan, Koh Murdar, Koh Maran) near Quetta.

Green House effect

Light rays from sun penetrates- glass of green house – absorbed by the plant and soil-reradiate as longer wave infra red radiation (heat). Glass does not permit these rays to escape outside so heat remains within the green house this increasing the temperature of earth.

Acid Rain

☐ pH of Normal rainwater is 5.3-6.0 .

☐ $\text{CO}_2 + \text{H}_2\text{O} = \text{H}_2\text{CO}_3$ (carbonic acid)

☐ When pH level of rainwater falls below this range due to the inclusion of carbonic acid, it becomes acid rain.

Pollution and it's types

Pollution is any undesirable change in physical, chemical or biological characteristics of air, land, water or soil. Agents that bring about such an undesirable change are called as pollutants.

Types:

☐ Air pollution

☐ water pollution

☐ land pollution

☐ light pollution

☐ litter

☐ Noise pollution

☐ Plastic pollution

☐ Soil contamination



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- ☐ Radioactive contamination
- ☐ Thermal pollution
- ☐ Visual pollution

Noise pollution

Noise is the unpleasant and undesirable sound which leads to discomfort in human beings.

- ☐ The intensity of sound is measured in decibels (dB).
- ☐ The faintest sound which can be heard by the Human ear is 1 Db.

Coniferous forests or tree

Coniferous forests are also known as Taiga or Boreal forests. They extend as a continuous belt across north America and north Eurasia below the arctic tundra .

These are characterized by conifers (gymnosperms). They are evergreen, drought resistant and woody. In many species the canopy is cone shaped.

The common species of trees of these forests are Spruce, fir and pine trees.

Sources of noise

- ☐ Natural sources
- ☐ Human-made sources

Natural Sources: volcanic eruption, the wild sea waves, a rush of rivers, thunder, roaring of animals, etc., They occur occasionally, but sometimes they are very damaging.

Man-made Sources: Man invented many machines for his convenience, which is now the primary cause of noise pollution.

Direct and indirect available resources

- ☐ Some resources, such as solar energy, fresh air, wind, fresh surface water, fertile soil, and wild edible plants, are directly available for use
- ☐ Other resources such as petroleum, iron, water found underground, and cultivated crops, are not directly available .

Marine Ecosystem Distribution :

Marine ecosystem covers nearly 71% of the earth's surface with an average depth of about 4000 m. Fresh water rivers eventually empty into ocean. Salinity of open sea is 3.6% is quite constant.

Sodium and chlorine-86% of the sea salt and the rest is due other elements such as sulphur, magnesium, potassium and calcium.

Climatic Change Health effects

- Heart problems
- Respiratory issues



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- Allergies
- Diseases spread
- Changing forest and natural areas
- Alteration in geographic distribution of vegetation type

Coniferous forests Flora and fauna :

These are characterized by conifers(gymnosperms). They are Evergreen, drought resistant and woody. In many species the canopy is cone shaped. The common species of trees of these forests are Spruce, fir and pine trees.

The productivity is much less than other ecosystem. There are very few animals in these forests. The herbivores are red squirrel, deer, goat, mule, moose etc.

Tropical grasslands Three strata

(i) Root layer-mostly root biomass-upper 16cm, roots may penetrate 1.7m down-underground stems or rhizomes

(ii) Ground layer-mosses and dandelions

(iii) Herbaceous layer-saesonlly short grasses, wild mustard, coneflower.

Sources of pollutants

The pollutants come from two types of sources.

1. Point sources

2. Non point sources

Point sources are single, identifiable sources.

Examples are,

- Smokestack of a coal-burning power or industrial plant,
- The drainpipe of a factory,
- The exhaust pipe of an auto mobile.

Nonpoint sources are dispersed and often difficult to identify.

Examples are pesticides blown from the land into the air and the runoff of fertilizers and pesticides from farmlands, lawns, gardens, and golf courses into streams and lakes. It is much easier and cheaper to identify and control or prevent pollution from point sources than from widely dispersed nonpoint sources.

Tropical grasslands Flora and fauna :

Grasses are the dominating plants with scattered drought resistant trees in the tropical grasslands. Trees are less than 10 m in height.

Animals-much reduced-because there is no shelter. The large herbivores of this biome are bison, proghorn (North America) wild horse, ass, saiga (Eurasia) , zebra and antelope (South Africa).



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Carnivores are quite small in number and size e.g: coyotes, weasels, badgers foxes and ferrets . Hawks, lark sparrows, warblers, Great Indian Bustard and peafowl are the common birds. Biomes has rich in reptilian and insect.

Radioactive contamination

“Radioactivity is a phenomenon of spontaneous emission of protons (alpha particles), electrons (beta particles) and gamma rays (short wave electromagnetic waves) due to the disintegration of atomic nuclei of some elements. These cause Radioactive Pollution”.

Solutions to Land Pollution

- ☐ Use of organic matures instead of artificial fertilizers in agriculture
- ☐ Planting of trees (Afforestation)
- ☐ Treating solid wastes such as garbage, domestic refuse, and industrial materials before dumping them in landfills
- ☐ Recovering and recycling of materials such as plastics, cloth, and glass before dumping.

Coniferous forests Climate :

Climate is cold. Long and harsh Winters is for more than six months. Mean annual temperature is below 0°C, Soil is poor in nutrients and acidic in nature.

Lake Water Pollution

Lakes are the rich source of water but due to following reasons, lake water gets polluted:

- i) Discharge of organic waste of homes and factories.
- ii) Discharge of organic matter of sewage treatment plants.
- iii) Dumping of garbage of villages and cities
- iv) Inorganic pollutants of fertilizers and pesticides used in agriculture

Effects of Air Pollution

- ☐ Smog
- ☐ Acid rain
- ☐ Eutrophication
- ☐ Global warming

Sources/Causes of light pollution

- ☐ Increased energy consumption through over-illumination from artificial light sources such as street lighting, domestic lighting, and garage
- ☐ Poor planning by engineers while placing street lights and signage

IDA

Bring Back the Dark Sky

- ☐ U.S.-based International Dark Sky Association (IDA), formed in 1988 to preserve the natural night sky



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- ☐ In 2017, the IDA approved the first U.S. dark sky reserve
- ☐ December of 2018, IDA lists thirteen dark sky reserves on their site.

Recycling and reusing

Reuse is using a resource over and over in the same form

- ☐ For example, glass bottles can be collected, washed, and refilled many times
- ☐ Recycling involves collecting waste materials and processing them into new materials
- ☐ For example, discarded aluminum cans can be crushed and melted to make new aluminum cans or other aluminum products

How ozone abundances are determined in stratosphere and troposphere?

Ozone abundances in the stratosphere and troposphere are determined by,

1. ▪ Amount of reactive gases
2. ▪ Sunlight intensity
3. ▪ Temperature

Causes of noise pollution

- ☐ Generators, mills, huge exhaust fans
- ☐ vehicles on the roads
- ☐ Weddings, fire works, public or Political or religious gatherings involve loudspeakers
- ☐ Mining, construction of buildings
- ☐ Noise from railway and aircraft

Interference competition:

Interference competition: occurs when an individual of one species directly interferes with an individual of another species.

Example: A plant releasing allelopathic chemicals to impede the growth of a competing species.

United Nations divided the world's countries in how many categories?

The United Nations classifies the world's countries into four categories.

1. The developed countries (with 1.2 billion people) include the United States, Canada, Japan, Australia, New Zealand, and most countries of Europe .
2. Most are highly industrialized and have a high per capita GDP PPP
3. All other nations (with 5.5 billion people) are classified as developing countries, most of them in Africa, Asia, and Latin America
4. Some are middle-income, moderately developed countries such as China, India, Brazil, Turkey, Thailand, and Mexico. Others are low-income, least developed countries where per capita GDP PPP is steadily declining.



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Prairie and Savanna

Grasslands are known by various names in different parts of the world.

For example they are called prairies in Canada and North America, steppes of Russia, Savannas in Africa and Pampas in Argentina.

The Prairies are known as the “Granaries of the world”. These are the extensive areas of flat or rolling grasslands.

Tropical grasslands are commonly called Savannas.

They occur in eastern Africa, South America, Australia and India.

Savannas-complex ecosystem-contain grasses with groups of trees. Soil of grassland is rich and fertile.

Acid Rain natural Cause

1. Natural sources

☐ Volcanic emissions

☐ Wildfires

☐ Lighting

☐ Decaying vegetation

Tundra Climate

A permanently frozen subsoil called permafrost is found in the arctic and antarctic tundra.

The summer temperature may be around 15°C and in winter, it may be as low as -57°C in arctic tundra .

A very low precipitation of less than 400 mm per year.

A short vegetation period of generally less than 50 days between spring and autumn frost Productivity is low.

Global warming

Global warming is the long-term warming of the planet's overall temperature. Though this warming trend has been going on for a long time, its pace has significantly increased in the last hundred years due to the burning of fossil fuels.

Climate of marine ecosystem

Temperatures in the ocean range from just around freezing at the pole and in the deep waters, to tropical clear waters that are as warm as a bathtub. The average temperature of all oceans is about 39°F (4°C). Heat from the sun warms only the surface of the water. Deep down, oceans everywhere are cold and dark.



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1. Groundwater Pollution

Water present below the surface of the earth is known as groundwater. More than 90% of freshwater is groundwater. Groundwater pollution occurs when harmful substances (pollutants) enter the groundwater. These pollutants are practically limitless but can range from motor oil to chemicals from agriculture to untreated waste.

Sources of water pollution:

Domestic waste

- Industrial waste
- Untreated sewage
- Oil spills
- Agricultural runoff
- Suspended or particulate matter
- Mining activities
- Acid rain

Salient features of Baluchistan desert

It has the world's largest deep sea port, the Port of Gwadar lying in the Arabian Sea. Although it makes up about 44% of the land area of Pakistan, only 5% of it is arable and it is noted for an extremely dry desert climate. Despite this, agriculture and livestock make up about 47% of Balochistan's economy.

Baluchistan lies in the desert belt between 25° N and 32° N. It has an arid to semi-arid climate, and water resources are limited compared to other provinces of Pakistan. The main sources of water on which people depend are springs, streams, rivers and karez. Usually associated with higher hills, stony plateaus. Very cold winters, no monsoon influence but occasional winter and spring showers.

Typified by Northern Kalat, lower parts of Suleiman hills, most of the Balochistan valleys, Kurram valley, most of Waziristan. Heavily overgrazed

Fauna: Hill fox, Balochistan race of Black bear, Hyena, Leopard, Marbled Pole Cat, caracal Cat, Urial, Goitred Gazella, Tibetan Hare, porcupine, Migratory Hamster, Balochistan Gerbil

Write the factors affecting the composition of communities and ecosystem..

The composition of communities and ecosystems changes due to these factors

1. Continents movement
2. Climate changes
3. Natural disturbances
4. Succession

Deserts fauna:

Desert animals are insects, reptiles, and burrowing rodents. Desert shrew, fox, kangaroo, wood rat, rabbit, armadillo are common mammals in desert. Camel is known as the ship of the desert



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Economic growth

Economic growth provides people with more goods and services, economic development has the goal of using economic growth to improve living standards. The United Nations classifies the world's countries as economically developed or developing based primarily on their degree of industrialization and their per capita GDP PPP.

The developed countries (with 1.2 billion people) include the United States, Canada, Japan, Australia, New Zealand, and most countries of Europe. Most are highly industrialized and have a high per capita GDP PPP.

All other nations (with 5.5 billion people) are classified as developing countries, most of them in Africa, Asia, and Latin America. Some are middle-income, moderately developed countries such as China, India, Brazil, Turkey, Thailand, and Mexico. Others are low-income, least developed countries where per capita GDP PPP is steadily declining.

Goals of environmental science

The three main goals of environmental science are: to learn how the natural world works, to understand how humans interact with the environment, and to find ways to deal with environmental problems and live more sustainably.

Predation.

The ecological process in which an animal (or an organism) kills and feeds on another animal (or an organism). The animal that kills another animal to feed on is called a "predator". The one that is killed to be eaten is known as prey.

Four levels of Biological organization

four levels of biological organization—organismal, population, community, and ecosystem. At the organismal level, ecologists study individual organisms and how they interact with their environments. At the population and community levels, ecologists explore, respectively, how a population of organisms changes over time and the ways in which that population interacts with other species in the community. Ecologists studying an ecosystem examine the living species (the biotic components) of the ecosystem as well as the nonliving portions (the abiotic components), such as air, water, and soil, of the environment.

Difference between spring and neap tides.

Spring tides occur twice each lunar month all year long without regard to the season. Neap tides, which also occur twice a month, happen when the sun and moon are at right angles to each other. The spring tides have higher high tides and lower low tides, whereas neap tides have lower high tides and higher low tides. The difference in the range of the water level between high and low tide is much larger in a spring tide than in a low tide.



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5. Sand Dunes and dunes found in Pakistan

- ☐ Extensive areas of undulating sand dunes, absence of cultivation and extensive semi desert.
- ☐ Thal desert, Thar and Cholistan desert, Nuski and Chagai sand dunes areas of Pakistan.

Characteristics of Tropical Rain (Evergreen) Forests :

Main characteristics

Temperature and light intensity very high

Rain fall is greater than 200 cm. per year.

Soil is rich in humus,

The rate of turnover of the nutrients is very high leading to high productivity and have highest standing crop and biomass

The vegetation includes broad evergreen trees of about 200 feet like

Health bamboos, ferns, shrub etc. Epiphytes and woody vines (liannas) are also abundant. Many tree species show buttresses (swollen stem bases) and leaves with drip tips.

Municipal solid waste

☐ Land pollutants are usually called MSW (Municipal Solid Waste).

It can be rubbish, trash, garbage, factories waste, food waste, plastic, and other materials. Pesticides, fertilizers, metals, lead, and hydrocarbons.

Litter and Garbage Household waste, broken items, etc Commercial establishments 240 million metric tones/year

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TAAHA Khan

Remember me in you Precious Prayer

Jazak Allah

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